

FIRST STEPS INTO THE OPERATION OF THE SAPOTI CRYOGENIC NANOPROBE AT THE CARNAÚBA BEAMLINE AT SIRIUS/LNLS

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Abstract

SAPOTI is the second nanoprobe at the CARNAÚBA (Coherent X-Ray Nanoprobe Beamline) beamline at the 4th-generation light source Sirius/LNLS. Working from 2.05 to 15 keV, it relies on simultaneous multi-analytical X-ray techniques (absorption, diffraction, spectroscopy, fluorescence and luminescence) and imaging in 2D and 3D. It has been designed for highly-stable fully-coherent beam sizes from 30 to 120 nm, and monochromatic flux up to 1×10^{11} ph/s/100 mA/0.01% BW after an achromatic KB (Kirkpatrick-Baez) focusing optics. Moreover, a new in-vacuum high-performance cryogenic sample stage has been developed aiming at single-nanometer resolution images. The nanoprobe is now successfully installed and technical commissioning is underway. The focus of this work is two-fold. Firstly, it highlights the system integration results at the beamline, namely: overall thermo-mechanical performance of the KB mirrors and sample stage. And, finally, it showcases the instrument's technical commissioning results, namely: KB alignment and focus stability, and initial fly-scan potential for ptychography and absorption imaging.

INTRODUCTION

Synchrotron scanning X-ray microscopy has evolved into a cornerstone characterization tool across scientific, technological, and engineering fields, apt at bridging the gap between conventional optical and high-resolution electron microscopy [1,2]. The pursuit of enhanced spatial resolution over the past decade has driven the development of numerous nanoprobe, which have pushed the boundaries of techniques like coherent X-ray diffractive imaging and ptychography to sub-5 nm resolution in 3D [3–9]. These advancements make the stability of the beam and the sample the ultimate limiting factors for resolution during the exposure time [10, 11].

SAPOTI, or Scanning Analysis by Ptycho for Tomographic Imaging, is the second nanoprobe to be deployed at the CARNAÚBA beamline at the Sirius light source, located at the Brazilian Synchrotron Light Laboratory (LNLS). Designed for a wide array of simultaneous multi-analytical X-ray techniques-including spectroscopy based on contrasts such as absorption, fluorescence, and luminescence, together with diffraction and coherent scattering imaging, such as Bragg CDI and ptychography-SAPOTI is a capable instrument for 2D and 3D imaging.

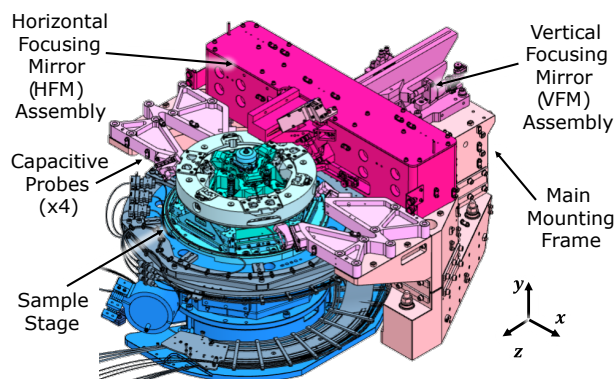


Figure 1: Drawing of the SAPOTI Sample Stage and KB module (see also Ref. [14]).

As detailed in [12, 13], achieving the highest imaging resolution required a tightly integrated mechatronic architecture, driven by the extreme sensitivity of the focusing optics (nanoradian scale) and the nanometer stability demanded by the sample. This work reports the first SAPOTI station results from the modules addressing these requirements—the sample manipulation stage and the KB mirror module Fig. 1. We summarize the system assembly, temperature control, and dynamical stability of these modules, along with the initial imaging results. The design and commissioning of high-performance controllers for nanopositioning scans will be presented in Ref. [15].

SYSTEM INTEGRATION AT THE BEAMLINE

Following the offline commissioning reported in Ref. [14], the station was transported to Sirius, where system integrity and functionality were verified. The mechatronic motion controllers, detector electronics, and advanced trigger system were integrated with EPICS [16] and the Bluesky orchestration environment [17]. Pulse-tube (PT) cryostats for both the loading module and sample manipulator were tested, yielding results consistent with Ref. [14]. Finally, modal tests in the hutch confirmed that the dynamical behaviour of the KB modules and sample manipulator met the required stability specifications [12, 13, 18].

Thermal deformation in synchrotron optics is a well-established research field, particularly for mirrors in 4th-generation light sources such as Sirius [19]. KB mirrors are especially susceptible to errors—such as coma aberration, astigmatism, and focus drift—if thermal expansion is

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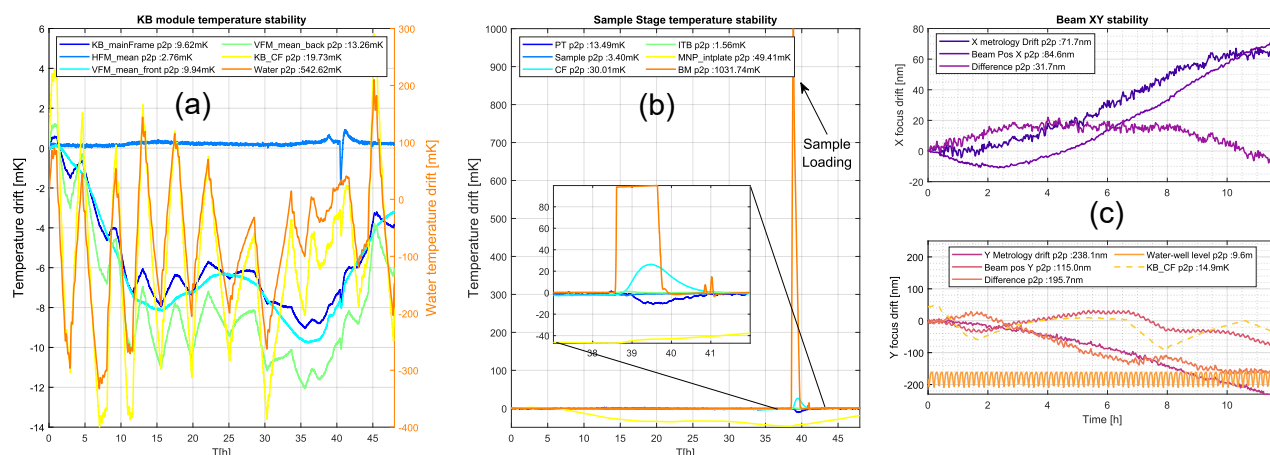


Figure 2: Thermal drift measurements of SAPOTI modules. (a) KB module thermal drift, (b) Sample Stage thermal drift, (c) XY measurement of focus drift compared with metrology measurements and temperature variation.

not carefully managed [20]. Likewise, precise temperature control of the Sample Stage is essential to ensure reliable metrology and to minimize sample-to-beam drift during long scans [12].

Both the KB module and the Sample Stage employed the in-house temperature-control architecture described in Ref. [21]. Figure 2(a) shows the KB module stability over 48 hours. While the process water varied by 540 mK, the closed-loop control of the KB cold finger (KB_CF), which couples the water to the back of the main mounting frame (see Fig. 1), attenuated fluctuations by nearly 30-fold. On the frame's front side, the HFM assembly (HFM_mean) showed only 2.7 mK variation. However, temperature gradients across the KB main frame—particularly between its front and back—can induce angular deformation in the Rx direction, leading to focused-beam drift along Y and displacement of the capacitive probes that complete the metrology chain between the KB module and the Sample Stage.

Figure 2(b) shows the Sample Stage temperature stability over 48 h during a simulated tomography experiment at room temperature (RT). In this setup, only the Sample Stage PT cooler was active to dissipate the heat from the voice-coil actuators [12]. During normal scanning, temperature oscillations remain below 6 mK. The zoomed view highlights the effect of sample loading, when the actuators operate beyond their nominal range and generate extra heat. Owing to the thermally decoupled architecture [18], this disturbance is limited to the Sample Stage cold finger (CF) and PT, which re-stabilize within about two hours. This decoupling minimizes thermally induced deformations on metrology parts, ensuring stability for long-duration experiments such as tomography scans.

Mechanical Stability

The preliminary results of the cryogenic sample stage, with a thorough discussion on the redesign of its dynamical and cryogenic parts, is detailed in Ref. [14]. In this work, it becomes possible to assess the actual performance met-

rics of real operational conditions in the hutch—namely, in vacuum, at cryogenic and room temperatures, and subject to environmental disturbances induced by the floor, vacuum pumps, and PT coolers.

Figure 3 shows the cumulative power spectrum (CPS) of stand-still stability measurements of the metrology chain that encompasses the Sample Stage and KB system. Root mean square values (σ) were considered in the 0.1 to 2 kHz range, peak to peak (p2p) considered the full range. Figure 3(a) shows the CPS of XYZ degrees of freedom (DoFs) of the short-stroke sub-module (Shs), measured via interferometers (IFM) with respect to the metrology frame (MFR). Vertical dashed lines are the harmonics (of 47 Hz) of the PT mechanism, which are the main contributions to vibrations in both RT and cryogenic sample operation.

In Fig. 3(b) is the CPS of the measurement of XYZ DoFs of the MFR with respect to KB mounting frame, measured via capacitive probes (CAP), in Abbé. The vibration level is under 3 nm. Furthermore, note that the PT's 3rd and 4th harmonics contribute the most to the MFR vibration. Contributions in the 200 to 500 Hz range are mainly due to internal dynamics of the sample stage [14]. Hence, using this measurement in image reconstruction may prove to be useful.

In Fig. 3(c) is the CPS of the ZRx DoFs for the VFM and RyRz DoFs for the HFM, measured via optical encoders. Note that system vibration is below the sensor noise level (smooth CPS curve characteristic), particularly in the frequency range of 0.1 to 1 kHz, which may indicate that no additional gain in image reconstruction can be obtained by using the KB encoders measurement.

SCIENTIFIC COMMISSIONING

Since the SAPOTI station shares beamtime with TARUMÁ [22], the scientific team had about two weeks so far for the scientific commissioning of the station and its detectors, with samples in RT for the sake of simplicity. Leveraging on the expertise gained on TARUMÁ, a focal spot of ≈ 40 nm was found via knife-edge fluorescence map-

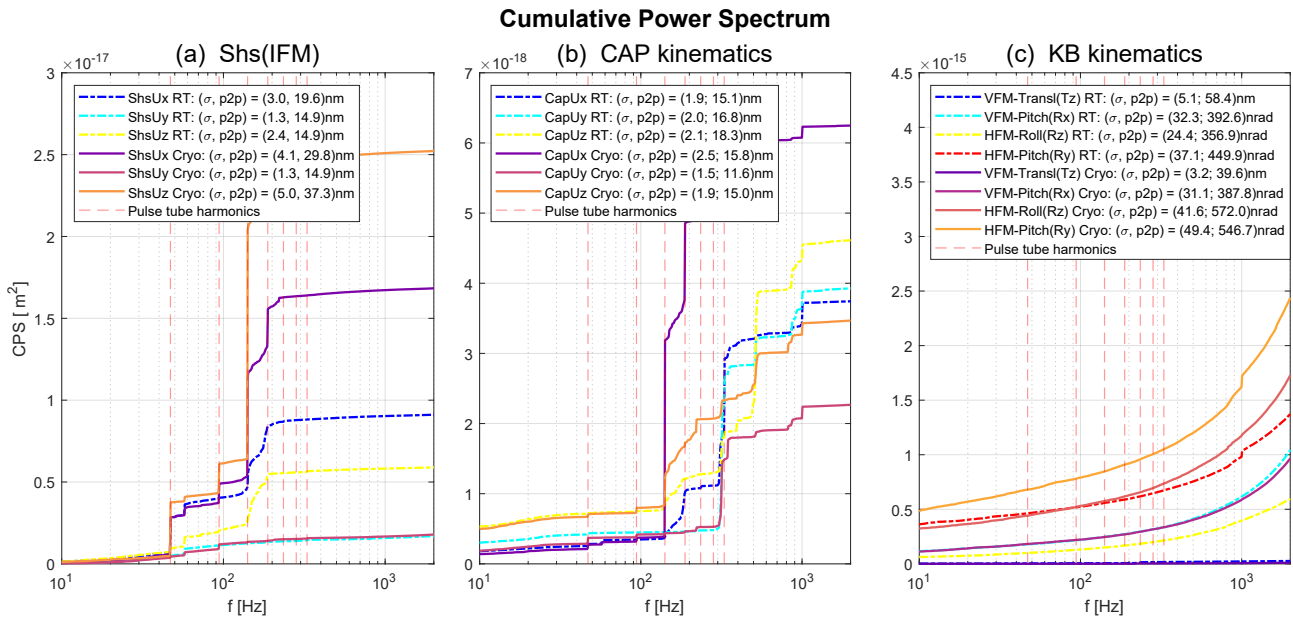


Figure 3: Cumulative power spectrum (CPS) of stand-still stability measurements for cryogenic and RT conditions. (a) Sample Stage Short stroke (Shs) (b) Sample Stage Metroframe (MFR) (c) KB mirrors, see also Ref. [12].

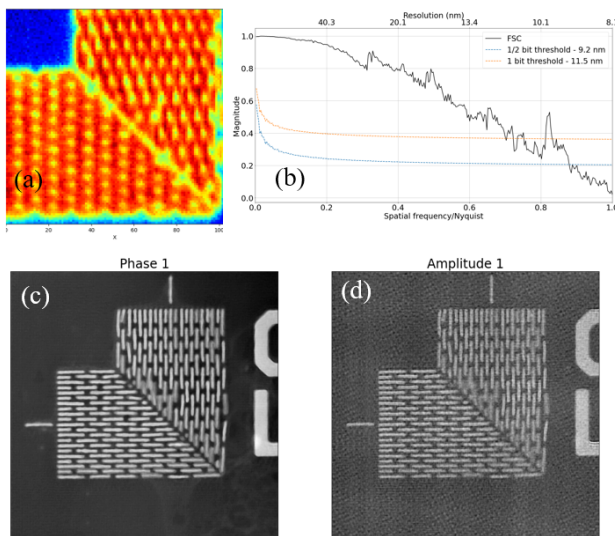


Figure 4: Applied Nanotools 50nm calibration patterns. (a) STXM. (b) FSC. (c) Ptycho phase and (d) amplitude.

ping and ptychographic reconstructions with a reference sample. Figure 4 shows the preliminary results obtained with X-rays at SAPOTI. From the STXM (scanning transmission X-ray microscopy) image in Fig. 4(a), it is possible to resolve patterns of 50 nm. In Figs. 4(c) and 4(d) one can observe the ptycho reconstruction of the same pattern and the Fourier shell correlation (FSC) graph (Fig. 4(b)) shows that the 1/2 bit threshold criteria yields 9.2 nm resolution.

In Fig. 2(c), a sequence of knife-edge scans measured the XY focus stability over 12 h. The relative metrology between the Sample Stage and the KB mirror is composed of the measurements of the capacitive probes and the KB encoders.

On the X and Y directions, the focus drifted around 70 nm and 115 nm, respectively. If metrology correction was to be applied, the drift would decrease to 30 nm on X direction whereas it would increase to 195 nm on the Y direction.

Moreover, Y focus drift curve has two interesting patterns: one low frequency (≈ 6 h), that follows the trend of the KB cold finger temperature (presumably due to thermal deformation of the KB main frame) and a 10 nm amplitude higher frequency one (≈ 10 min) that matches the CNPEM water-well dynamics. This effect can be traced back from the parasitic movement of VFM's primary source which is the undulator. From the sensitivity analysis done in Ref. [20], one can see that the primary source should move about 1.0 to 1.5 μ m to yield a 10 nm effect on the VFM focus.

CONCLUSION

Although commissioning is still ongoing, the preliminary results indicate that single-nanometer resolution imaging may be realistically achieved in the near future. To this end, efforts will be made to optimize both the use of metrology and the image reconstruction algorithms. The next steps focus on validating tomographic experiments with cryogenic samples. In parallel, the implementation of fluorescence and luminescence techniques will be validated, further extending the station's experimental capabilities.

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